1999 IEEE International Fuzzy Systems Conference Proceedings
August 22-25, 1999, Seoul, Korea

Fuzzy Logic and Intelligent Agents

John Yen, Magy Seif El-Nasr, Thomas R. Ioerger
Center for Fuzzy Logic, Robotics, and Intelligent Systems
Department of Computer Science
Texas A&M University
College Station, TX 77843-3112
{yen,magys,ioerger}@cs.tamu.edu

1. Introduction

Intelligent agents are software modules that can interact with its external environment in an autonomous fashion to achieve its goals. Multiple distributed agents can communicate to collaborate or to compete. Agents often can learn from feedback in an incremental fashion. During the past few years, there has been an increasing interest in the technology and the applications of intelligent agents. Applications of intelligent agents range from NASA's Deep Space Exploration, meta-search engines on the Web, to E-commerce.

One of the major issues faced by all types of agents are incomplete information about the external environment. A mobile robot agent, for instance, may not have a complete detailed map of its surroundings. Furthermore, most agents interact with an environment that is dynamic (i.e., it changes over time). For example, an agent in intelligent networks needs to consider the dynamic network traffic conditions. Whether an agent uses fuzzy logic or not, it has to deal with these issues regarding uncertainty. Fuzzy logic can play a major role in an intelligent agent for reasoning about the dynamic environment (Yen and Langari, 1999).

Furthermore, fuzzy logic can also be used to express an agent's goals that can be partially satisfied. In fact, long before the agent paradigm became popular, fuzzy logic had already been successfully used in representing ``fuzzy goals''. One of the most well known examples of such is Hitachi's Sendai Subway control system developed by S. Yasunobu in the 80's (Yasunobu and Miyamoto 1985). The Sendai Subway control system uses fuzzy goals to evaluate alternative control decisions. The use of fuzzy goals enables an agent to maximize its overall satisfaction degree by considering options that partially satisfies each individual goal. For example, this is especially useful when an agent needs to deal with multiple goals that are potentially conflicting in nature.

2. A Fuzzy Logic-based Agent: FLAME

FLAME (Fuzzy Logic Adaptive Model of Emotions)is a new computational model of emotions that can be incorporated into intelligent agents and other complex, interactive programs (El-Nasr, Yen, and Ioerger 99). The model uses a fuzzy-logic representation to map events and observations to emotional states. The model also includes several inductive learning algorithms for learning patterns of events, associations among objects, and expectations. We demonstrated empirically through a computer simulation of a pet that the adaptive components of the model are crucial to users' assessments of the believability of the agent's interactions.

Emotions are an important aspect of human intelligence and have been shown to play a significant role in the human decision-making process. Researchers in areas such as cognitive science, philosophy, and artificial intelligence have proposed a variety of models of emotions. Most of the previous models focus on an agent's reactive behavior, for which they often generate emotions according to static rules or pre-determined domain knowledge. However, throughout the history of research on emotions, memory and experience have been emphasized to have a major influence on the emotional process.

FLAME is based on a psychology model about emotions developed by Roseman (Roseman 1990), which generates emotions according to an event assessment procedure. The model divided events into motive-consistent and motive-inconsistent events. Motive-consistent events are events that are consistent with one of the subject's goals. On the other hand, a motive-inconsistent event refers to an event that threatens one of the subject's goals. The notion of ``motive'' in Roseman's model is similar to the notion of ``goals'' in intelligent agents. In FLAME, the black-and-white event desirability assessment in Roseman's model (i.e., classifying an event into motive-consistent vs motive-inconsistent) is generalized into one that allows event desirability to be a matter of degree. In another word, the goals of an agent is fuzzy in that an event can partially contribute to a goal.

FLAME consists of three major components: an emotional component, a learning component and a decision-making component. The agent first perceives external events in the environment. These perceptions are then passed to both the emotional component and the learning component. The emotional component will process the perceptions; in addition, it will use some of the outcomes of the learning component, including expectations and event-goal associations, to produce an emotional behavior. The

0-7803-5406-0/99/$10.00 ©1999 IEEE

behavior is then returned back to the decision-making component to choose an action. The decision is made according to the situation, the agent's mood, the emotional states and the emotional behavior; an action is then triggered accordingly.

The emotional component of FLAME first evaluates an external event using two steps. First, the experience model determines which goals are affected by the event and the degree of impact that the event holds on these goals. Second, mapping rules compute a desirability level of the event according to the impact calculated by the first step and the importance of the goals involved. The event evaluation process depends on two major criteria: the importance of the goals affected by the event, and the degree by which the event affects these goals. Fuzzy rules are used to determine the desirability of an event according to these two criteria. Considering an agent that interacts with a user as a ``virtual pet'', one of the goal of the agent is to avoid hunger. If an event has a negative impact on the goal (e.g., if the food dish is taken away by the user), it may trigger the following rule in evaluating the event:

IF Impact (prevent_starvation, e) is HighlyNegative
AND Importance(prevent_starvation) is
VeryHigh
THEN Desirability(e) is HighlyNegative

where e denotes an external event being evaluated.

FLAME uses fuzzy sets to represent emotions, and fuzzy rules to represent mappings from events to emotions, and from emotions to behaviors. Fuzzy logic provides an expressive language for working with both quantitative and qualitative (i.e., linguistic) descriptions of the model, and enables our model to produce some complex emotional states and behaviors. For example, the model is capable of handling goals of intermediate importance, and the partial impact of various events on multiple goals. Additionally, the model can manage problems of conflicts in mixtures of emotions (Elliot 1992). Though these problems can be addressed using other approaches, such as functional or interval-based mappings (Velasquez 1997, Reilly 1996), we chose fuzzy logic as a formalism mainly due the simplicity and the ease of understanding linguistic rules. We will describe below the fuzzy logic models used in FLAME.

Fuzzy logic is used to determine a behavior based on a set of emotions. The behavior depends on the agent's emotional state and the situation or the event that occurred. For instance, the following rule describes how a pet will behavior under a mixture of anger and fear emotions.

IF Anger is *HighIntensity*
AND Fear is *MeduimIntensity*
AND Event is *dish-was-taken-away*
THEN BEHAVIOR is growl.

Based on FLAME, we developed an interactive simulation of a pet called PETEEI — a PET with Evolving Emotional Intelligence. We evaluated the system by designing a questionnaire for uses to answer after they went through several pre-defined scenarios for interacting with the pet.
We asked the user to evaluate four versions of PETEEI: one without fuzzy and learning, one with fuzzy logic but not learning, and one with both fuzzy logic and learning. While fuzzy logic did not contribute significantly in the user evaluations, it provided a better means of modeling emotions due to its qualitative and quantitative expressiveness. Learning, on the other hand, was found to be most important for creating the appearance of intelligence and believability. Even though the pet was not animated, the users all agreed that the model behind the interface could serve further as a basis to simulate more believable characters. As a matter of fact, one of our users noted that, "I like this model better than Petz [a commercial product that simulates believable pets], because this model shows me more about how the pet learns and displays his affections to me."

3. Conclusion

Even though we have shown how fuzzy logic can be used to deal with various issues regarding uncertainty for an intelligent agent. Much more remains to be explored. Some of the agent-related issues that fuzzy logic can contribute to include multi-agent negotiation, theory refinement under uncertainty, and the use of fuzzy information retrieval techniques in personalized Web-based information broker agents.

4. References

[1] C. Elliot. The Affective Reasoner: A Process model of emotions in a multi-agent system. Institute for the Learning Sciences, Northwestern University, Ph.D. Thesis, 1992.

[2] M. Seif El-Nasr, J. Yen, T. R. Ioerger, FLAME - Fuzzy Logic Adaptive Model of Emotions, *International Journal of Autonomous Agents and Multi-Agent Systems,* in press, 1999.

[3] J. Roseman, P. E. Jose, and M. S. Spindel. Appraisals of Emotion-Eliciting Events: Testing a Theory of Discrete Emotions. *Journal of Personality and Social Psychology*, 59 (5), 899-915, 1990.

[4] J. Velasquez. Modeling Emotions and Other Motivations in Synthetic Agents. *Proceedings of the AAAI Conference 1997*, Providence, RI, 10-15, 1997.

[5] S. Yasunobu and S. Miyamoto, Automatic train operation by fuzzy predictive control. In M. Sugeno ed. *Industrial Applications of Fuzzy Control*, North Holland, 1985.

[6] J. Yen and R. Langari, *Fuzzy Logic: Intelligence, Control, and Information*, Prentice Hall, 1999.